



EtabsX

BNBC 2020 workflow assistant for ETABS

User Manual · version 2.0

TwinAnalytic · twinanalytic.com/etabsx
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What this program is

EtabsX reads your open ETABS model over the CSI API, computes the loads BNBC 2020 asks for, writes them back into the model, and then checks the analysed result against the Code. Every number it prints carries the clause it came from.

It is an assistant for the repetitive half of the work. The engineering judgement, the design decisions and the professional seal remain yours.

What it will not do, stated first

- **It never saves your model.** Your undo is closing ETABS without saving.
- **It does not run the analysis.** You press F5 in ETABS yourself.
- **It does not design the members.** ETABS does that; EtabsX reads the results and applies the BNBC floors to them.
- **It does not choose R for you.** It tells you what the wall share implies.
- **It does not replace your engineering judgement, or your seal.**

Where this manual and the printed Code disagree, **the Code is right.**

Licence

There is one EtabsX. Not a trial edition and a full edition — this program, which runs **free for 30 days** on a computer and afterwards asks for a licence key that unlocks the same executable in place. No second download, no reinstall, nothing lost.

How a licence is obtained

1. Open EtabsX. The sidebar shows a **machine code** like
ABCD-1234-EF56-7890. Copy it — do not retype it.
2. Send it with the plan you want, at
<https://twinanalytic.com/etabsx>.
3. You receive a key. Paste it into the licence box and carry on.

Plans

Plan	Length
Project	30 days
Quarterly	90 days
Annual	365 days
Perpetual	no expiry

A key is bound to **one computer**. The machine code comes from the Windows installation, so it survives renaming the PC, joining a domain and reinstalling EtabsX — but not reinstalling Windows, which is a new machine and needs a new key.

While a licence is running

The sidebar states where the copy stands and changes colour as the end approaches: gold while there is time, amber inside two weeks, red in the last three days. A term licence is re-checked **every launch** against its own expiry date, so the countdown is the licence, not a reminder.

Verification is **offline**. The program never contacts a server to run, so it works on a site machine with no internet.

If a key is cancelled

A licence can be revoked — a refunded payment, or a key issued in error. The program notices the next time it starts and asks for a new key. You are told at once and can finish and save the session you are in; nothing is taken away mid-work.

The check sends nothing. A short public list of cancelled-key fingerprints is downloaded and compared on your own machine, so the program still never reports who is running it. If it cannot reach the list — offline, firewalled, or the file is missing — nothing happens at all.

System requirements

Operating system	Windows 10 or 11, 64-bit
ETABS	ETABS 22, installed and registered, running with your model open
Install	None. A single executable, with Python bundled.
Optional	openpyxl for native .xlsx export — without it every export falls back to CSV automatically

ETABS must run at the same elevation as EtabsX — both as administrator, or neither. Mixed elevation is the single most common reason the connection silently attaches to nothing.

The Seismic and Guide tabs compute without ETABS. Everything else reads from or writes to a live model.

Installing and launching

There is no installer. Download EtabsX.exe and double-click it.

Windows will warn that the publisher is unknown, because the file is not code-signed. Choose **More info** → **Run anyway**. The SHA-256 of every release is published beside the download if you would rather verify it first.

The first launch takes a few seconds — a one-file build unpacks itself to a temporary folder — and shows briefly as **two processes**, a small loader and the real application. Only the second owns the window.

What it stores on your computer

About 650 bytes, in five places, so that deleting one does not lose your licence:

```
%LOCALAPPDATA%\TwinAnalytic\.axetabs.dat
%APPDATA%\TwinAnalytic\.axetabs.dat
%PROGRAMDATA%\TwinAnalytic\.axetabs.dat
a hidden file beside EtabsX.exe
HKCU\Software\TwinAnalytic\EtabsX
```

Each holds two timestamps and a flag, signed so it cannot be hand-edited. No name, no company, no computer name, no model data.

Your last session's inputs are also kept, so the fields are where you left them next time:

```
%LOCALAPPDATA%\TwinAnalytic\EtabsX\last-session.json
```

Nothing is transmitted. The only network traffic the program makes is a plain GET for the version file and the cancelled-key list, and neither sends anything about you.

Connecting to ETABS

1. Open ETABS and load your model. Leave it open.
2. Open EtabsX.
3. Press **Connect** in any tab that needs the model.

If the connection fails, in order of likelihood:

1. **Elevation mismatch** — see System requirements above.
2. **ETABS is not running**, or has no model open.
3. **More than one ETABS instance** is running; close the extras.
4. **The model is mid-analysis**. Wait for it to finish.

The status line names the model file when it attaches. If it names a different file from the one you are looking at, you have two instances.

The safety model

The program is built so that a mistake in it cannot cost you a day's work.

- **It never saves.** Every write goes into the in-memory model only. Close ETABS without saving and everything EtabsX did is gone.
- **Writes are announced before they happen.** Each writing action states what it is about to change and how many objects it will touch.
- **Reads are the default.** The Audit, Checks and Results tabs cannot alter anything.
- **The Fix buttons re-run the check** they just applied, so a correction that did not work says so immediately.

The recommended habit is the obvious one: save your model yourself, under a new name, before the first write.

The order of work

The tabs are arranged left to right in the order you use them. Going through them in sequence is a complete pass.

You leave the program twice, both times deliberately:

```
IN ETABSX tabs 1 - 9 compute the loads and write them in
|
IN ETABS run the analysis (F5), then the concrete design
|
IN ETABSX tabs 10 - 16 audit, check, report, detail
```

After the first pass, stop working sequentially. Open **Compliance**, press **Run all checks**, and work from the findings: each carries a **Fix** that applies the correction and re-runs it, and a **Go** that opens the tab owning the detailed check. That is how design actually goes — from the failures, not from the top.

Tab reference

Seventeen tabs. The subtitle under each name in the program is repeated here so the manual and the window agree.

Every tab follows the same shape: inputs on the left, a log pane on the right, a Preview before any write. Clause references are shown on-screen next to each value.

How to read the figures. Every control discussed is boxed and numbered, and the colour tells you what it does before you press it:

Colour	Meaning
Blue	An input you fill in, or a read-only view. Safe.
Amber	Reads the model and changes nothing. Safe.
Red	Writes to the ETABS model. Preview first.

There is usually only **one red control per tab**. If you press nothing red, the program cannot alter your model.

Tab 1 — Seismic

BNBC 2020 Sec 2.5.7 — equivalent static force. **Works offline:** everything except ■ and ■ runs with no model open.

Why this tab exists. ETABS 22 has no BNBC auto-seismic option. Left to itself you would hand-translate the BNBC spectrum into ASCE 7-05 boxes, and a slip anywhere in that chain is silent — the model runs, the numbers are just wrong. This tab does the translation, shows the clause beside every value, and refuses the combinations the code does not permit.

When to use it. First, before any load pattern exists. Everything downstream — combinations, drift limits, the P-Δ check — depends on SDS and R being right, and they are decided here.

#	Control	What it does	When you touch it
1	Tab rail	The workflow runs top to bottom. Tab 15 Guide prints the same order.	Follow it in order the first time
2	Connection state	Grey dot = offline. Tabs 1, 2 and 15 still work; everything that reads or writes a model needs Connect .	Only for ■ and ■
3	Location → Seismic zone	Table 6.2.15 town lookup. Sets the zone only , and deliberately leaves every other box alone, so a deliberate override is never undone behind your back.	Always — then check the zone it chose
4	Occupancy category	Table 6.2.17 importance factor I. Also selects which half of Table 6.2.18 the SDC comes from — Occupancy IV reads the right-hand half and lands one category more severe in most cells.	Always

#	Control	What it does	When you touch it
5	System group → System	Table 6.2.19 gives R , Ω and C_d . Changing the group resets the system to the first in that group.	Always — the most consequential input on the tab
6	Clause column	The BNBC table or equation behind each row.	When checking someone's work
7	Height h_n	Height above the seismic base, for T_a . Excludes the basement where its walls tie to the ground slab — Sec 2.5.7.2(b).	Always
8	Computed T (optional)	Blank = use T_a . Enter a modal period and it is capped at $1.40 \times T_a$ — the code will not let you claim the model's full flexibility.	After the first analysis
9	Seismic base storey	Grade level. Storeys below it are excluded from ΣW and take no force.	Set it when there is a basement
10	Apply X / Y force to pattern	The user-force patterns ■ writes into, created if missing. Keep them separate from the auto EQ_x/EQ_y so you can compare the two.	Leave at EU_x / EU_y
11	Calculate	Runs the whole clause chain and prints it. Changes nothing, needs no model.	Every time an input changes
12	Read T from model	Pulls the largest modal period into ⑧. Needs the analysis to have been run.	After the first analysis
13	Export to Excel	The full calculation as a sheet, for the design file.	When issuing
14	Apply seismic forces to ETABS	The only control here that writes. Distributes $V = S_a \cdot W$ over the storeys (Eq 6.2.41) and applies it as user joint loads on the diaphragms.	Once mass and diaphragms are real
15	Log pane	Every result, clause and warning. This is the output — read it, don't just look for a number.	Always

How to use it

1. Set ③ **Location**, then confirm the **Seismic zone** it picked.
2. Set ④ **Occupancy**, ⑤ **System group / System**, ⑦ **Height h_n** and ⑨

Seismic base storey.

3. Press ■ **Calculate**. Read the log: SDC, the system check, S_a , and the ASCE 7-05 values to type into ETABS.
4. Model, mesh and run the analysis. Come back, press ■ **Read T from model**, then ■ **Calculate** again — the period is now the model's, capped at $1.40 \cdot T_a$.

- Only then consider ■.

What it gives you to type into ETABS. ETABS 22 has no BNBC auto-seismic, so use **ASCE 7-05** with $SDS = (5/3) \cdot Z \cdot S$, $SD1 = (5/3) \cdot Z \cdot S \cdot TC$, Site Class **F (user defined)**, and the **Fa** and **Fv** the tab prints.

Before you press ■. It applies **user joint loads on the diaphragms**, not an auto-seismic pattern. That is deliberate: this build's API will not set auto-seismic parameters, so writing the forces directly is the only honest way to get the BNBC distribution into the model. It needs four things and says so if any is missing:

- **Calculate** run first — it needs S_a and T .
- A **connected model** with storeys.
- Assembled joint mass** — set the Mass Source (Tab 6) and run the analysis

once, so ETABS has actually assembled the mass.

- Rigid diaphragms** assigned (Tab 5), or there are no joints to load.

The log then prints, per storey, the force and how many nodes took it. **Check that node count.** A storey with one node is a storey whose diaphragm did not take.

Traps

- hn **excludes the basement** when basement walls tie to the ground slab — the building is fixed at grade, not at the raft.

- EQx must be **X only** and EQy **Y only**. One pattern doing both halves the force in each direction.

- I/R > 1.0 is refused** — Sec 2.5.4.3 does not permit it, and the log says so in red rather than quietly carrying on.

- S_a has a **floor**. Where the decayed spectrum falls below $C \cdot \beta \cdot Z \cdot I$ the floor governs; the log prints both so you can see which one you are on. Tall, high-R buildings live on the floor more often than people expect.

- R , Ω ■, C_d and SDS in *Concrete Frame Design* ■ *Preferences* must match the load pattern. They do not update themselves.

- 5% accidental eccentricity is required (Sec 2.5.7.6.2).

Tab 2 — Wind

BNBC 2020 Sec 2.4 — analytical procedure.

Why this tab exists. The BNBC wind chapter is ASCE 7-05 with a different exposure alphabet and a different basic-speed map. The two most expensive mistakes in Bangladeshi wind design — typing a m/s speed into an mph field, and reading the exposure letter as if it were ASCE — both happen in the translation. This tab keeps the units and the letters straight.

When to use it. After the storey list exists in the model, because the per-storey forces need real heights.

#	Control	What it does	When you touch it
1	Tab rail	Wind is step 17 of the order of work.	—
2	Connection state	■ and ■ need a live model.	—
3	Location → Basic wind speed V	Table 6.2.8 town lookup, in m/s , 3-second gust, 50-year, Exposure B.	Always — then sanity-check the speed

#	Control	What it does	When you touch it
4	Exposure (BNBC)	Sec 2.4.6.3. BNBC A/B/C are not ASCE A/B/C — see the conversion below.	Always
5	Occupancy category	Table 6.2.9 importance factor lw. Occupancy I drops to 0.77 above 44 m/s.	Always
6	Kzt, Kd	Topographic factor (1.0 on flat ground) and directionality, Table 6.2.12.	Rarely
7	Gust factor G	Leave blank to compute it. Typing 0.85 on a flexible building is a code violation the log will call out.	Leave blank
8	Fundamental period T■	Above 1.0 s the building is flexible and Sec 2.4.8.2 makes the computed Gf mandatory.	After the analysis
9	Damping β	Fraction of critical: 0.01 concrete, 0.02 steel.	Set by material
10	Plan dimension X, Y	For the gust factor.	Always
11	Parapet height	0 if none.	If there is one
12	Wind starts at storey	Grade level — not Base if there is a basement.	With a basement
13	Apply X / Y force to pattern	User-force patterns, kept separate from the auto wx/wy.	Leave at WUx / WUy
14	Clause column	The clause behind every row.	—
15	Read storeys from model	Pulls the storey list and heights. Reads only.	First, every time
16	Calculate	Velocity pressure, gust factor and per-storey forces.	After ■
17	Export to Excel	The full calculation.	When issuing
18	Apply wind forces to ETABS	The only control here that writes.	Last
19	Log pane	The storey table and every warning.	Always

How to use it

1. ■ **Read storeys from model** — nothing else is meaningful without heights.
2. Set ③ **Location**, ④ **Exposure**, ⑤ **Occupancy**, ⑩ **plan dimensions**.
3. Leave ⑦ **Gust factor blank**. Press ■ **Calculate**.
4. After the analysis, put the real T■ in ⑥ and calculate again.

Traps

- **Exposure letters do not match ASCE.** BNBC A → ASCE B (urban),

BNBC B → ASCE C (open), BNBC C → ASCE D (open water). Getting this wrong changes the pressure by tens of percent.

- **Units.** Dhaka is $V = 65.7 \text{ m/s} = 147 \text{ mph}$. Typing 65.7 into an mph field

gives you about a fifth of the correct load.

- **Flexible buildings.** $T_1 > 1.0 \text{ s} \Rightarrow$ the rigid $G = 0.85$ is not permitted. If you force it anyway the log says so explicitly.
- With no period entered the tab **assumes rigid** and warns you it did.

Tab 3 — Sections

Beam and column library plus materials.

Why this tab exists. Typing forty rectangular sections into ETABS by hand is an hour of clicking and a reliable source of typos. This builds the whole library from a text list.

When to use it. Early, right after grids and storeys — before you draw anything you want to assign a section to.

#	Control	What it does	When you touch it
1	Tab rail	Step 2 of the order of work.	—
2	Connection state	The whole tab needs a live model.	—
3	Column / Beam material	Read from the model, or pick a preset. A different grade per member type is fine and normal.	Always
4	Read materials from model	Lists what the model already defines. Reads only.	First
5	Create standard materials	Writes. Creates C3000 / C4000 concrete and 500W steel only if the model does not already define them — your own materials are never touched.	Once per model
6	Size units	mm or inch. The lists are read in whatever you pick here.	Set once
7	Column list	width x depth (T2 x T3), one per line.	Always
8	Beam list	Same format.	Always
9	Preview	Says exactly what would be created, and what already exists.	Before ⑩
10	Create sections in model	Writes. Creates every section in both lists.	After ⑨
11	Log pane	What was created and what was skipped.	Always

Traps

- The format is **width x depth**, i.e. T2 x T3. 300x600 is a 300 wide, 600 deep beam — not the other way round.
- Existing sections are **skipped, not overwritten**. If you want to change a size, change it in ETABS.
- Set ⑥ **before** typing the lists, or you will create 250 mm columns that were

meant to be 250 inch.

Tab 4 — Patterns

Load patterns in the model.

Why this tab exists. Two things go wrong with load patterns and both are invisible afterwards: self-weight counted twice, and a seismic pattern that acts in both directions at once. This tab makes both visible before they get in.

When to use it. Step 13 — after geometry and assignments, before area loads.

#	Control	What it does	When you touch it
1	Tab rail	—	—
2	Connection state	Needed for both buttons.	—
3	The pattern list	Name, type, self-weight multiplier and purpose. Untick use to leave a pattern out. Edit any name to match your own convention.	Always
4	Self wt column	Exactly one pattern may carry 1.0 — normally Dead. Every other row must be 0.0 or the structure's own weight is counted twice.	Check every time
5	Purpose column	Why each pattern exists, with its clause. PW carries the Sec 2.3.6 partition allowance — minimum 1.0 kPa if the partitions are movable.	Read it once
6	Check model	Compares the list against the model and reports what is missing. Reads only.	First
7	Create missing	Writes. Creates only what is absent; existing patterns are untouched.	After ⑥
8	Log pane	What exists, what is missing, what was created.	Always

Traps

- **Self-weight double counting.** If both Dead and, say, FF carry 1.0, the building weighs twice what it should — and the seismic base shear with it. The Audit tab checks this too, because it is that easy to get wrong.
- EQ_x/EQ_y and w_x/w_y must be **direction-specific**. One pattern doing X and Y halves the force in each.
- H is type **Other** — there is no earth-pressure pattern type in ETABS.

Tab 5 — Loads

Area loads by storey.

Why this tab exists. Area loads are per-storey data entry, which is where transcription errors live. Reading the storeys from the model, filling from an occupancy preset and assigning in one pass removes almost all of them.

When to use it. Step 14, after the patterns exist.

#	Control	What it does	When you touch it
1	Tab rail	—	—
2	Connection state	Whole tab needs a live model.	—
3	Read storeys · Fill down from top row · units · Preview	Reads the storey list, replicates the top row down the building, sets psf or kPa, and shows what would be assigned. All read-only.	Always, in that order
4	ASSIGN to model	Writes. Applies every area load in the grid.	Last
5	also assign a rigid diaphragm per storey	Writes a diaphragm as well. Leave it ticked — the seismic and wind force application both need diaphragm joints to load.	Leave ticked
6	Occupancy preset	Fills the whole grid with typical values for that occupancy. A starting point, not an answer.	Then edit
7	Storey column	Read from the model, with its elevation. The top row is the roof.	—
8	The load grid	FF, PW, Live, Lr per storey.	Always
9	Log pane	What was read and what was assigned.	Always

Typical BNBC 2020 residential values (psf) — printed on-screen:

Load	Value	Reference
FF floor finish	25 – 30	—
PW partition walls	20 – 40	Sec 2.3.6
Live residential	40	Table 6.2.3
Live corridor / stair / balcony	100	Table 6.2.3
Lr accessible roof	100	—
Lr non-accessible roof	20	—

Traps

- **Do NOT enter slab self weight.** ETABS computes it from the Dead pattern's self-weight multiplier. Adding it here counts the slab twice.
- The roof row defaults to **Lr only** — change it if your roof also carries a floor live load.
- **Fill down from top row** overwrites every row below. Fill first, then edit the exceptions.

Tab 6 — Assign

Stiffness modifiers, mass source, restraints and mesh — four sub-tabs.

This is the tab that decides whether your analysis means anything. All four sub-tabs write to the model.

6a — Stiffness modifiers

#	Control	What it does	When you touch it
1	Tab rail	Step 9.	—
2	Sub-tab strip	Four separate jobs; do them left to right.	—
3	Frames	Beam I33 = I22 = 0.35 (cracked, ACI 318 6.6.3.1.1(a)); beam torsion J = 0.10, which releases the compatibility torsion; column I33 = 0.70. Frame area/shear stays 1.0.	Always
4	Walls	F11 = F22 = F12 = 0.35 is what governs drift — the membrane terms, not m11. M11 matters for basement walls retaining earth.	Always — get this right
5	Slabs	M11 = 0.25 where the beams take the flange, else 0.35. F11 = 1.0 — the diaphragm must stay stiff in plane.	Always
6	Read what is in the model	Reports the modifiers already assigned. Reads only.	First
7	APPLY to all members	Writes to every member of each type.	After ®
8	Log pane	—	—

The trap that costs the most. Cracking a shear wall means reducing the **membrane** terms f11, f22, f12. Changing only m11/m22 does almost nothing, because a shear wall works in-plane. A model with f11 = 1.0 walls is far too stiff, and every drift and period you read from it is wrong.

6b — Mass source

#	Control	What it does	When you touch it
1	Tab rail	Step 10.	—
2	Sub-tab strip	—	—
3	Multiplier per pattern	Dead, FF and PW are always 1.0 (PW by Sec 2.3.6). Live is 0 for residential, 0.25 for storage . Lr is 0 unless snow > 1.5 kPa.	Always
4	also include element self mass	Leave OFF — it double counts , because the Dead pattern already carries self weight at 1.0.	Leave off
5	lateral / vertical mass	Lateral ON . Vertical OFF unless you are checking Ev by modal means.	Leave as set

#	Control	What it does	When you touch it
6	Read current	Reports the model's current mass source. Reads only.	First
7	SET mass source	Writes.	After ⑥
8	Log pane	—	—

Why this matters more in Bangladesh than in an ASCE workflow. BNBC includes a **fraction of the floor live load** in seismic mass (25% / 50% by occupancy); ASCE 7 does not. An engineer trained on ASCE will leave Live at 0 by habit and under-estimate the seismic mass of a storage or assembly building.

6c — Supports and mesh

#	Control	What it does	When you touch it
1	Tab rail	Steps 7 and 8.	—
2	Sub-tab strip	—	—
3	Base level tolerance	Joints within this of the lowest z count as base joints.	Rarely
4	Restraint	Fixed for a raft or tied pad footings; Pinned for isolated pads on soft soil. It changes the whole building's period and drift.	Choose deliberately
5	Preview base joints	Lists what would be restrained. Reads only.	First
6	APPLY restraints	Writes.	After ⑤
7	Auto edge constraint ON for all areas	Writes. Ties a mesh node landing mid-edge of a neighbouring panel back into that panel.	Always, once
8	The rigid-zone note	Rigid zones over walls are a separate ETABS switch . Turning them on shortens the clear span of beams framing into wide walls — correct for wide piers, optimistic for narrow ones.	Read once
9	Log pane	—	—

Without auto edge constraint, a wall and the slab beside it can share an edge and **transfer nothing**. The model runs, looks fine, and is wrong.

6d — Geometry doctor

#	Control	What it does	When you touch it
1	Tab rail	Step 6.	—
2	Sub-tab strip	—	—
3	Near-miss tolerance	Joints this close but not merged.	Widen it if you suspect sloppy geometry
4	Diagnose	Finds and lists the bad joints. Reads only.	First

#	Control	What it does	When you touch it
5	Select the bad joints in ETABS	Selects them in the model so you can see them. Still changes nothing.	Then
6	What to do by hand	Edit ■ Merge Joints, Edit ■ Align Joints/Frames/Edges (Shift+Ctrl+M), Draw ■ Edit lines / Divide.	Follow it
7	Log pane	—	—

It finds it, shows it, and does not silently move it. ETABS 22's EditPoint interface has no working move method through the API, and moving joints by script is how models get destroyed. Fix one cluster by hand, re-run the diagnosis, repeat.

Tab 7 — Earth H

At-rest lateral earth pressure on basement walls.

#	Control	What it does	When you touch it
1	Tab rail	Step 15.	—
2	Connection state	Ⓔ and ■ need a live model.	—
3	Retained height, water table depth	Grade down to the underside of the raft; water table from your geotechnical report .	Always
4	ϕ , γ , γ_{sat}	$K_{\blacksquare} = 1 - \sin \phi$. Unit weight above water, saturated below.	From the soil report
5	Surcharge q	Traffic, an adjacent footing, construction plant.	Always consider it
6	Load pattern	Type Other — ETABS has no earth-pressure type.	Leave at H
7	Basement storeys	Comma separated, exactly as named in the model.	Always
8	Seismic increment	Mononobe-Okabe, $\Delta\sigma = 0.375 \cdot kh \cdot \gamma \cdot H$ with $kh = SDS/2.5$.	Tick in SDC C and D
9	Calculate	Pressure diagram and totals. Changes nothing.	First
10	Preview panels	Which wall panels would be loaded. Reads only.	Then
11	APPLY to walls	Writes.	Last
12	Log pane	—	—

Traps

- The **water table dominates**. Below it you carry both the buoyant soil pressure and the full hydrostatic pressure; getting the depth wrong changes the total by a factor, not a percentage.
- **At rest, not active**. A basement wall braced by floors cannot rotate enough

to mobilise active pressure. K_a , not K_a .

- The pressure **sign is guessed from panel geometry** — check the deformed shape for a non-rectangular basement.

Tab 8 — Combos

BNBC 2020 Sec 2.7 — strength and service combinations.

Why this tab exists. You need **two** families of combinations for two different jobs, and using the wrong one is a silent, expensive error: factored loads oversize a footing by roughly 50%, service loads under-reinforce the concrete.

#	Control	What it does	When you touch it
1	Tab rail	Step 18.	—
2	Connection state	Needed to create.	—
3	SDS, redundancy ρ , overstrength Ω	Pull SDS from the Seismic tab with ⑦. Leave ρ at 1.0 — BNBC applies ρ on drift (Sec 2.5.14.1), not on E.	Always
4	Wind / H / live factors	1.6 on service wind (Sec 2.7.3.1); 1.3 permitted for RC by Exception 3. Live factor 0.5 permitted where $L \leq 5$ kN/m ² , except garages and assembly.	Check against your case
5	Pattern names	Must match the names in the model exactly. Dead patterns are comma separated and all share the dead factor.	Always
6	The option ticks	Orthogonal 100 % + 30 % (required in SDC C and D); include H; the H-absent twins of combos 6 and 7; SERVICE (ASD) for footing sizing; OVERSTRENGTH for collectors; SERVICEABILITY for wind drift; ENV-ALL; and whether to register them as concrete design combinations.	Read every one
7	Pull SDS from Seismic	Copies SDS across so the two tabs cannot disagree. Reads only.	First
8	Preview	Prints every combination it would build.	Then
9	CREATE in model	Writes.	After ⑧
10	Audit existing	Compares what is in the model against the BNBC set. Reads only.	On an inherited model
11	Log pane	Every combination, with its clause.	Always

The two families

Family	Prefix	What it is for
STRENGTH (Sec 2.7.3.1)	UD- UW- UE-	Sizing reinforcement, checking the concrete
SERVICE / ASD (Sec 2.7.2.1)	SD- SW- SE-	Footing plan area against allowable bearing; settlement, overturning, sliding

Service combinations are deliberately **not registered** as concrete design combinations, so ETABS cannot accidentally design to them.

Traps

- Using **factored** loads to size a footing oversizes it by about 50%. Using **service** loads to design the footing concrete under-reinforces it. You need both sets in the model.
- Uplift almost always appears under the **0.9D** combinations. That is exactly what they are for — it is not a modelling error.
- Overstrength combos are built but **not auto-registered**; they apply to collectors and elements supporting discontinuous walls, and choosing which members is your job.

Tab 9 — Piers

Pier and spandrel labelling.

#	Control	What it does	When you touch it
1	Tab rail	Step 12.	—
2	Connection state	—	—
3	Endpoint tolerance	Panels whose ends fall within this are treated as one pier.	Widen for sloppy geometry
4	Label prefix	P1, P2, ... in order of discovery.	Leave at P
5	same label on every storey	Leave ticked — a pier is a vertical stack, not one panel.	Leave ticked
6	Preview grouping	Shows which panels would share a label. Reads only.	First
7	ASSIGN labels	Writes.	After ⑥
8	List existing	What is already labelled. Reads only.	On an inherited model
9	Log pane	—	—

Why it matters. A wall panel without a pier label produces **no wall design and no pier force output**. That is why the wall-share check on the Results tab comes back empty on an unlabelled model. Spandrels — the strip of wall over an opening — get their own labels only if you have openings; a solid wall does not need them.

Tab 10 — Audit

Ten read-only model checks. Changes nothing, ever.

When to use it. Step 19 — before you press F5, every time. Two minutes here saves hours later.

#	Control	What it does
1	Tab rail	—

#	Control	What it does
2	RUN ALL CHECKS	Runs all ten. A progress card names the check being run. Reads only.
3	The report	Ten numbered sections, in the order you should fix them.
4	The verdict tag	[PASS] green, [CHECK] amber, [FAIL] red. Work the reds first.
5	The → advice lines	Under every finding: what it means and what to do about it.

What it looks at

#	Check
1	Load patterns and self-weight double counting
2	Mass source (Sec 2.5.7.3 — dead + partitions, not live)
3	Load combinations against BNBC Sec 2.7.3.1
4	Base restraints
5	Diaphragm assignment per storey
6	Stiffness modifiers, cracked section (ACI 318 6.6.3.1.1)
7	Auto edge constraint / mesh continuity
8	Wall panels without pier labels
9	Orphan joints and unsupported beam ends
10	Slab / wall / frame section counts

Read section 9 first. "250 joints belong to nothing" and "18 beam ends land on nothing" mean the model is not connected — a beam end that touches no column, wall joint or other beam is a cantilever the model does not know about. Nothing below that matters until it is clean.

Tab 11 — Checks

Irregularity, Ax, θ and response-spectrum scaling. Post-analysis, read-only.

#	Control	What it does
1	Tab rail	Step 22.
2	Cd, I, R and the case names	Must match the Seismic tab and the model's case names.
3	RUN ALL	Runs every check. Reads only.
4	The report	One section per check, with the clause.
5	Per-storey ratios	The whole height, not just the worst value — so you can see where the problem is.
6	The verdict and its advice	What the number means and how to fix it.

What gets checked

Tag	Check	Threshold
H1	Torsional irregularity	max/avg storey drift > 1.2 / 1.4
H2	Re-entrant corners	projection > 15% of the plan
H3	Diaphragm discontinuity	area drop > 50%
V1a	Soft storey	stiffness < 70% of the one above
V2	Mass irregularity	mass > 150% of an adjacent storey
V3	Vertical geometric	dimension > 130% of adjacent
Ax	Accidental torsion amplification	Sec 2.5.7.6.2
θ	P- Δ stability coefficient	Sec 2.5.14.4
—	Storey force distribution	$F_x = V \cdot w_x \cdot h_x^k / \Sigma$, Sec 2.5.7.6
—	Response-spectrum scaling	to 85% of the static base shear

How to read a torsion failure. Over 1.2 is torsional irregularity and Ax amplification applies. Over 1.4 is **extreme** — not permitted in SDC E/F, and in SDC D the diaphragm and collector forces must be amplified 25%. The fix is to **move** stiffness toward the flexible side, not to add it symmetrically.

Tab 12 — Results

Modal, drift, reactions and wall share. Post-analysis, read-only.

#	Control	What it does
1	Tab rail	Step 21 — run this before Tab 11.
2	Structure type, occupancy, Cd, I	Table 6.2.21 is a table, not a number: the structure row and the occupancy column decide the drift limit. Set them before reading a drift verdict.
3	The report	Six reports in order.

The six reports, in order

1. Modal periods and mass participation — the 90% rule, Sec 2.5.11.
2. Whether **mode 1 is torsional** — if it is, fix the plan before anything else.
3. Base reactions per load case.
4. Design storey drift $\Delta = C_d \cdot \delta / I$ against the limit.
5. Wall share of base shear → whether the R you claimed is justified.
6. Concrete frame and shear wall design pass/fail.

The wall-share report is the honesty check on R. You chose R on the Seismic tab by naming a system. This tells you what the model actually behaves like. If you claimed a shear-wall R and the walls carry a third of the shear, the R is not justified and every force in the building is too low.

Tab 13 — Compliance

The BNBC checklist — check, fix, and see what is next.

Why this tab exists. It is the same body of checks as Audit, Checks and Results, re-ordered into **the order of work** and given a fix button. It answers the question the other tabs do not: *what should I do next?*

#	Control	What it does
1	Tab rail	Use it whenever you lose your place.

#	Control	What it does
2	Run all checks	Runs the whole list. A progress card names the check being run. Reads only.
3	Storey profile	Per-storey drift and torsion, top to bottom, so you can see where along the height a problem is.
4	The counts	n pass · n fix · n look.
5	The checklist	Three groups: MODEL READINESS (do these in order), ANALYSIS CHECKS (evaluated from results), ANALYSIS REVIEW (open the tab for the full check).
6	Log pane	The summary, and the detail for whichever row you press ? on.

The verdicts

Chip	Meaning
PASS green	Nothing to do.
CHECK amber	Look at it — the program cannot decide for you.
FAIL red	Fix it.
NA grey	Not applicable yet — usually a prerequisite is missing.
GO blue	Opens the tab that owns the detailed check.

Each row can carry three buttons on the right: [Fix] applies the correction and re-runs the list, [Go] opens the tab that owns the check, and [?] prints the full explanation into the log pane.

Work top-down. Fix the first red item, then run the checks again. The order of the list is the order of the work.

Tab 14 — Output

Reports, quantities, the preliminary footing sizer, and the project file.

#	Control	What it does	When you touch it
1	Tab rail	Steps 24–26.	—
2	Connection state	Everything except Save/Load inputs needs a live model.	—
3	Foundation reactions → Excel	Every restrained joint, every combination: FZ, FX, FY, MX, MY.	For the footing designer
4	Quantity take-off → Excel	Concrete volume and formwork area per storey, split into columns, beams, walls and slabs.	For the estimator
5	Member schedule → Excel	Every column, beam, wall and slab section with its storey and its length or area.	For the drawings
6	Design-basis report → HTML	One page with every input, clause reference and check result. Opens in a browser; print it to PDF.	For the design file

#	Control	What it does	When you touch it
7	Footing sizer inputs	Allowable bearing from the soil report; overhead ≈ 1.10 for the pad's own weight and backfill; $f'c$; the assumed square column size for the punching perimeter; and the maximum L/B a rectangular pad may reach.	Before ⑨
8	rectangular pad where $e > B/6$	With this ticked, an eccentric pad grows its long side along the moment until the whole base bears, instead of growing a square in both directions.	Leave ticked
9	Size isolated footings	One pad per column from the SERVICE reactions, with punching and one-way shear depth and rebar both ways.	After the analysis
10	Save / Load inputs	Every box on every tab, to a .aex file, so the next building starts where this one finished.	End of a project
11	Log pane	The footing schedule and every warning.	Always

What the footing sizer does. Plan area comes from the **service** envelope, $e = M/P$, against your allowable bearing. While $e \leq L/6$ the pressure is the trapezoid $q = P/A \cdot (1 + 6e/L)$; past that the base lifts and the partial-contact form $q = 2P/(3B(L/2 - e))$ governs — the trapezoid **under-predicts** it, so both are in there. Depth is the larger of two-way punching (perimeter at $d/2$) and **one-way (beam) shear** at d from the column face, both on the factored peak pressure. The sheet says which one governed each pad, the bearing pressure it actually reaches, and — where the base still lifts — how much of it is left in contact with the soil.

It is preliminary, and here is exactly how. Isolated pads on an assumed square column; the resultant moment is taken about one axis. **Biaxial moment, combined / strap / mat footings, sliding and settlement are not checked.** It also puts a pad on **every restrained joint**, so a meshed shear-wall base produces one small nonsense pad per node rather than one wall footing, and it does not check whether two pads overlap on plan. Do the final foundation design in SAFE.

Traps

- Size the pads with the **SERVICE** (SD/SW/SE) rows and design the concrete with the **STRENGTH** (UD/UW/UE) rows. If the service set is missing, the tab says so and falls back conservatively.
- A joint in **net uplift** needs a tie or an anchor, not a bearing pad.
- On granular soil, allowable bearing is itself **width-dependent** — a value valid for a 1.5 m pad is not valid for the 4 m pad the tool just sized.

Tab 15 — Report

One PDF: what was checked, against which clause, with what result — and, as loudly, what could not be checked.

Why this tab exists. Every verdict this program reaches lives in a log pane and disappears when you close the tab. The document you hand to a client, a reviewer or an approving authority has to survive that, and it has to be honest about its own gaps.

Why it is a tab and not a button on Output. A report has preconditions. They belong in front of you *before* the document is made, not discovered in a finished PDF that has already been emailed.

#	Control	What it does	When you touch it
1	Check what is ready	Reports what the PDF <i>would</i> contain — how many findings are FAIL, CHECK, NA and PASS, and exactly what the incomplete banner will say. Changes nothing.	Before generating
2	Generate PDF	Writes the report and opens it. Re-runs the compliance checks and the Checks tab first, so the document reflects the model as it is now rather than whatever was last on screen.	When ready
3	Log pane	Where the file went, and whether it was marked incomplete.	Always

The title block comes from the Detailing tab — project, client, job number, revision, prepared by, checked by. One place to set it, so the drawings and the report cannot disagree.

What is in it

1. **The model** — storeys, joints, columns, beams, wall panels, slabs
2. **Design basis** — the seismic parameters and the drift limit, each with its clause
3. **Compliance checks** — every dashboard check, worst first
4. **Results** — torsion, soft storey and the modal table, each with the measurement it rests on
5. **Irregularity and detailing checks** — the Checks tab's findings
6. **Not checked by the software** — the register of what you must verify by hand

How it refuses to mislead you.

- **NA is printed, never omitted, and sorts above PASS.** A check that could not run is not a pass, and it does not get to sink to the bottom of the document.
- **Every number carries its provenance** — which case, which load step, measured on what. The soft-storey row prints " $k = V/\text{drift}$, storey shears from Story Forces, softest of 2 load step(s)" rather than a bare number.
- **A report missing its inputs says so on every page.** If the seismic or wind parameters have not been calculated, the footer reads **INCOMPLETE — DO NOT ISSUE**, the filename carries (INCOMPLETE), and the closing statement says which sections are missing.
- **It certifies nothing.** There is a signature block for the engineer of record and a line saying in as many words that the software does not certify the document.

Tab 16 — Detailing

*AutoCAD DXF drawings and the material take-off, straight from the analysis and the concrete design. **New in 2.0.***

Why this tab exists. The model already knows every span, every section and every required steel area. Retyping that into a CAD package is a day of work and a fresh chance to make a transcription error. This produces the drawings and the quantities from the same numbers the analysis used.

When to use it. Last. It needs the analysis **and** the concrete design to have been run in ETABS. Without design results it falls back to code-minimum steel and stamps the sheet *CODE MINIMUM — design not run*, which is honest but is not a design.

> **Two boxes are blank on purpose and block every drawing button** — the > **seismic detailing class** and the **coarse aggregate** (rows 1a and 1b > below). They are engineering decisions the program will not make for you, so > the first drawing you attempt will stop and ask. Set both before you start and > you will not meet them one at a time.

#	Control	What it does	When you touch it
1	Tab rail	2.0 has seventeen pages; Detailing is the last one that produces anything.	—
1a	Seismic detailing class — <i>REQUIRED, blank on purpose</i>	SMF, IMF or OMF. No drawing is produced until this is chosen. It sets the tie spacing (Sec 8.1.9 / 8.3.10.5 / 8.3.5.4(f)) and whether laps may sit at a support (Sec 8.3.4.2(b)) — the program will not guess it, because the answer follows from your structural system and Table 6.2.19, not from anything in the model. See Tab 1 — Seismic : the R you took there already implies this, and the Compliance tab checks the two agree.	First, every time
1b	Coarse aggregate — <i>REQUIRED, blank on purpose</i>	Stone or Brick. No drawing is produced until this is chosen. Sec 8.1.11.2(b) wants 1.5× the shrinkage steel in brick-aggregate concrete, which is common in Bangladesh and cannot be inferred from an ETABS model.	First, every time
1c	Project · Client · Job number · Revision · Drawn by · Checked by	The title block on every sheet — and the Report tab's header, so the drawings and the report cannot disagree. Left blank, sheets show a dash rather than a guess.	Once per job
2	Units	mm (metric) or ft-in (imperial). Imperial prints every dimension as ft-in, names the bars #3, #4, #5 ..., takes f'c in psi and fy in ksi, and reports steel in lb. Switching resets the boxes below so none is left holding a millimetre number under an inches label.	Set it first

#	Control	What it does	When you touch it
3	Clear cover · Bar sizes available	Cover is to the outside of the link. The bar list is what your supplier stocks; longitudinal steel takes 16 then 20 first (imperial #5 then #6) and only goes bigger when the preferred size will not fit in one layer.	Always
4	Link diameter from · Max link spacing	A starting size — links step up only if the spacing would fall below 75 mm. Spacing is rounded down to a 25 mm module.	Rarely
5	Rebar fy · Wastage and offcuts · Concrete f'c	f'c sets the lap and development lengths. Laps are drawn and counted, so the wastage figure is offcuts only.	Always
6	Sheet size · Plot scale	The drawing is built for this sheet. Text and dimensions plot at 2.5 mm at this scale, whatever scale you choose.	Set once
7	Beam detailing → DXF	Continuous runs, curtailment, laps, zoned links, two section cuts, bar bending schedule.	After the design
8	Column detailing → DXF	Elevation with splices and confinement, plus a 1:20 section with crossties.	After the design
9	Slab detailing → DXF	Panel plan with marks and thickness, plus a typical section with both mats drawn.	After the design
10	Shear wall detailing → DXF	Pier elevation and a section through the boundary zone. Needs pier labels — unlabelled panels are skipped and reported.	After the design
11	Total material take-off → Excel	Concrete, formwork and rebar. Reads the bar bending schedules the four drawings produced, so the tonnage and the drawings cannot disagree.	Last
12	Log pane	What was read, what was drawn, and every warning.	Always

These write nothing to your model. Every button on this tab reads the model and writes a file. That is why none of them is drawn as a primary action.

What the drawings actually contain

	Detail
Beams	Collinear beams sharing a joint are drawn as one continuous run . Top steel curtailed at $\max(0.25 L, l_d)$ each side of every support and dimensioned; bottom steel lapped over interior supports and hooked at end supports; two continuous top bars; links tighter over $2h$ from each face.
Columns	Stacks lift by lift, with the confinement zone $l_o = \max(h/6, \text{larger dimension}, 450)$ at both ends, the tighter tie spacing inside it, and the splice in the middle half. Crossties to Sec 8.1.9 wherever the bar pitch needs them.
Slabs	Panel plan per level, plus a typical section per thickness with both mats drawn at their real spacing.
Walls	Pier elevation, and a horizontal section showing the boundary zone with its links, both curtains of the web, and a break line.

The code it follows

Quantity	Clause
Development length l_d	BNBC 2020 Sec 8.2.3.2 — and $\sqrt{f'_c}$ is capped at 8.3 MPa, Sec 8.2.2
Standard hook l_{dh}	Sec 8.2.6.2, not less than $8d$ nor 150 mm
Tension lap	Class B, Sec 8.2.13.2 (1.3 l_d). Bars 36 mm and larger may not be lapped — Sec 8.2.12.2(a)
Bar spacing in a layer	Sec 8.1.6 — clear spacing $\geq \max(25 \text{ mm}, d, 4/3 \text{ aggregate})$
Column ties, crossties	Sec 8.1.9 — every corner and alternate bar held, no bar $> 150 \text{ mm}$ clear from one
Confinement inside l_o	Sec 8.3.5 — $s \leq \min(b/4, 6d, s_{max})$, $h_x \leq 350 \text{ mm}$
Slab and wall distributed steel	Sec 8.1.11 — spacing capped at the lesser of $5h$ and 450 mm

Development length assumes the unfavourable case unless it can prove otherwise. Sec 8.2.3.2 has two columns; the favourable one needs cover and clear spacing to meet limits, which is checked from the cover you typed and the spacing the bars actually land at. Where it cannot be shown, the longer length is used — a T20 top bar goes from 941 mm to 1454 mm.

Where it stops

- **Slabs carry shrinkage and temperature steel only.** Flexural design, column and middle strips, punching at the columns and openings are **not** in the drawing. Design the slabs in SAFE.
- **Wall boundary elements are sized on geometry** ($0.01 A_g$ over the zone), not on demand. Whether a special boundary element is required at all is a check this does not make.
- Beam-column joint ties, strong column–weak beam, and coupling beams are not detailed.
- The schedule uses BS 8666 shape codes for the shapes it draws, and nothing else.

****General note 11 on every sheet reads: *generated from the analysis model, to be checked and signed by the engineer of record before construction.* Leave it there.****

Tab 17 — Guide

The code rules this program follows.

#	Control	What it does
1	Tab rail	Available offline, always.
2	Section list	Fourteen sections, from the order of work to what the program does not do.
3	The text	The clause, the rule, and the common mistake, with ! marking the traps.

Section 0 is the order of work — the same sequence as section 8 of this manual, and the authoritative on-screen copy. When this manual and the in-app Guide disagree, the Guide is the one that ships with the software.

The Code rules this program follows

This chapter is the **Guide** tab, reproduced. It is read from the program at build time, so the manual and the window cannot disagree.

0. Order of work

Do it in this order. Each step depends on the one before it, and doing them out of order is what produces a model that quietly gives wrong answers.

#GEOMETRY FIRST

```
1 Grids and storeys. Basement storey included, with its real height.
2 Materials and sections. f'c, fy, and the real member sizes.
3 Draw columns, then beams, then walls, then slabs.
4 Merge coincident joints (Edit > Merge Joints).
5 Align beams to wall edges (Edit > Align Joints/Frames/Edges, Shift+Ctrl+M).
!6 Nothing below matters until every beam end lands on a column, a wall
! joint or another beam. Run the Model audit tab and clear section 9.
```

#ASSIGNMENTS -> tab 5 Assign

```
7 Base restraints on every joint at the lowest level.
8 Auto edge constraint on all slabs and walls.
9 Stiffness modifiers (cracked section).
10 Mass source.
11 Diaphragms, one per storey -> tab 4 Loads.
12 Pier labels on every wall panel -> tab 8 Piers.
```

#LOADS

```
13 Load patterns -> tab 3 Patterns.
14 Area loads per storey -> tab 4 Loads.
15 Earth pressure on the basement walls -> tab 6 Earth H.
16 Seismic parameters -> tab 1, then enter them in the auto load.
17 Wind parameters -> tab 2, then enter them in the auto load.
18 Load combinations, strength AND service -> tab 7 Combos.
```

#BEFORE YOU RUN

```
19 tab 9 Audit. Clear every FAIL. This takes two minutes and saves hours.
```

#RUN AND CHECK

```
20 Run the analysis in ETABS (F5).
21 tab 11 Results - modal, base reactions, drift, wall share, design.
22 tab 10 Checks - irregularity, P-Delta, Ax, spectrum scaling.
23 Fix what fails, re-run. Change ONE thing at a time.
!24 If you change several things at once and the model gets worse, you
! will not know which change did it. This costs hours.
```

#WHEN IT PASSES -> tab 12 Output

```
25 Foundation reactions to Excel, for the footing designer.
26 Quantity take-off and member schedule.
27 Design-basis report, and save the inputs as a .aex project file.
```

1. Seismic — Sec 2.5

#WHAT BNBC GIVES YOU

Z Table 6.2.14 zone coefficient. Dhaka is Zone 2, $Z = 0.20$.

S,TB,TC,TD Table 6.2.16 from the site class.

I Table 6.2.17 importance. Residential is II, $I = 1.00$.

R, Ω ,Cd Table 6.2.19 from the structural system you actually built.

Ct,m Table 6.2.20 for the approximate period.

#THE CALCULATION

$T_a = C_t \cdot (h_n)^m$ Eq 6.2.38

$T = \min(\text{computed } T, 1.40 \cdot T_a)$ Sec 2.5.7.2(a)

Cs = the normalised spectrum, branch by T against TB, TC, TD

$S_a = (2/3) \cdot Z \cdot I \cdot C_s / R$ Eq 6.2.34

$V = S_a \cdot W$

!hn EXCLUDES the basement when the basement walls are tied to the ground

!slab – the building is considered fixed at grade, not at the raft.

#WHAT ETABS WANTS

ETABS 22 has no BNBC auto-seismic. Use ASCE 7-05 and feed it parameters that reproduce the BNBC spectrum exactly:

$SDS = (5/3) \cdot Z \cdot S$ $SD1 = (5/3) \cdot Z \cdot S \cdot TC$

Site Class = F (user defined), then enter Fa and Fv from this program.

+The Seismic tab prints every one of these ready to type in.

#TRAPS

!EQx must be X direction ONLY and EQy Y ONLY. One pattern doing both

!halves your forces in each direction.

!The R, Ω , Cd and SDS in Design > Concrete Frame Design > View/Revise

!Preferences must match the load pattern. They do not update themselves.

!Accidental eccentricity 5% is required – Sec 2.5.7.6.2.

2. Wind — Sec 2.4

#THE LETTERS ARE NOT THE SAME

!BNBC Exposure A = ASCE Exposure B (urban, closely spaced)

!BNBC Exposure B = ASCE Exposure C (open terrain)

!BNBC Exposure C = ASCE Exposure D (flat, open water)

Choosing "A" in the ETABS ASCE dialog when you meant BNBC A gives you open-water pressures on an urban site. This is the single most common wind error in a BNBC model.

#THE CALCULATION

$K_z = 2.01 \cdot (z/z_g)^{(2/\alpha)}$ $z \geq 4.57$ m Sec 2.4.6.6

$= 2.01 \cdot (4.57/z_g)^{(2/\alpha)}$ below that

$q_z = 0.000613 \cdot K_z \cdot K_{zt} \cdot K_d \cdot V^2 \cdot I$ kN/m², V in m/s Eq 6.2.17

$p = q \cdot G \cdot C_p$

C_p windward = +0.80, leeward from L/B (–0.5 to –0.2)

Storey force $F = p_{net} \cdot B \cdot \text{tributary height}$

#UNITS

!V for Dhaka is 65.7 m/s. In an ETABS field labelled mph that is 147.

!Typing 65.7 into a mph field understates the wind by a factor of five.

#WHAT ETABS WANTS

Exposure from: DIAPHRAGMS. Assign diaphragms before you define the wind pattern or ETABS has no area to push on and returns nothing.

Create two patterns, Wx at 0° and Wy at 90°.

+Check the base reaction afterwards. Wx must produce FX, not FY.

3. Load combinations — Sec 2.7.3.1

#THE SEVEN BASIC COMBINATIONS

1 $1.4(D + F)$

2 $1.2(D + F + T) + 1.6(L + H) + 0.5(L_r \text{ or } R)$

3 $1.2D + 1.6(L_r \text{ or } R) + (1.0L \text{ or } 0.8W)$

4 $1.2D + 1.6W + 1.0L + 0.5(L_r \text{ or } R)$

5 $1.2D + 1.0E + 1.0L$

6 $0.9D + 1.6W + 1.6H$

7 $0.9D + 1.0E + 1.6H$

!H appears in combinations 2, 6 and 7 ONLY. Putting the earth pressure
!in all of them is wrong and unconservative in the ones where H resists.

#THE VERTICAL SEISMIC COMPONENT

$E = p \cdot QE \pm Ev$ where $Ev = 0.2 \cdot SDS \cdot D$.

Because Ev is proportional to D , it is folded into the dead factor:

combination 5 $\rightarrow (1.2 + 0.2 \cdot SDS) \cdot D + 1.0L + pQE$

combination 7 $\rightarrow (0.9 - 0.2 \cdot SDS) \cdot D + pQE$

For $SDS = 0.45$ that is $1.29D$ and $0.81D$.

+That is why the generated combinations show 1.29 and 0.81 and not $1.2/0.9$.

#ORTHOGONAL EFFECTS

In Seismic Design Category C and D, apply 100% in one direction with 30%
in the other, both signs, both orders. That is 8 permutations, applied to
combinations 5 and 7, giving 16 seismic combinations.

#WIND SIGN

Wind must be applied both ways: $+W_x$, $-W_x$, $+W_y$, $-W_y$. Twelve wind
combinations in total from basic 3, 4 and 6.

#TOTAL

3 gravity + 12 wind + 16 seismic = 31 strength combinations.

+The Combinations tab builds exactly these and can audit what is already
+in the model against them, factor by factor.

4. Mass source and seismic weight — Sec 2.5.7.3

Seismic weight W is the DEAD load plus:

- the actual partition weight, or 0.5 kPa minimum where partitions
can be moved (Sec 2.3.6)
- 25% of the floor live load in storage and warehouse occupancies
- the full weight of permanent equipment
- flat-roof snow where it exceeds 1.5 kPa

For a residential building NO floor live load enters W .

#IN ETABS

Define > Mass Source > Modify:

Mass from LOADS only (not "element + additional")

Dead 1.0 FF 1.0 PW 1.0 Live 0

✓ Include Lateral Mass

X Include Vertical Mass

!"Element self mass and additional masses" double counts the slab if you
!have also put the slab weight on as a load.

Read the resulting W in the Auto Seismic Load Calculation table after the
analysis, and compare against your hand figure: floor area $\times$ (slab weight
+ FF + PW) + column, beam and wall self weight. They should agree within
about 1%.

5. Stiffness modifiers — ACI 318-19 Table 6.6.3.1.1

#THE NUMBERS

beams $I_{33} = 0.35 I_g$

columns $I_{33}, I_{22} = 0.70 I_g$

walls cracked 0.35

walls uncracked 0.70

flat plates / slabs $0.25 I_g$

#THE TRAP THAT COSTS PEOPLE DAYS

A shear wall resists lateral load IN ITS OWN PLANE. In an ETABS shell that
is the MEMBRANE stiffness: f_{11} , f_{22} , f_{12} .

The bending terms m_{11} , m_{22} , m_{12} govern out-of-plane behaviour only.

!Setting $m_{11}/m_{22} = 0.35$ on a shear wall and leaving f_{11}/f_{22} at 1.0 does

!almost nothing to the drift. Set the f -terms.

For a basement wall retaining earth, the out-of-plane m -terms matter too,
so set both.

#SLAB MODIFIER AND DOUBLE COUNTING

If you switch on insertion-point stiffness transformation so the beams act as T-beams with the slab, the slab flange is already being counted in the beam. Drop the slab m11/m22 from 0.35 to 0.25 or you count it twice.

#WHEN TO USE 0.70 ON WALLS

Only if the wall genuinely stays uncracked under the design event. For a building in SDC D, assume 0.35. It gives a longer period, larger drift and a smaller base shear – check drift with 0.35 and strength with both.

6. Drift — Sec 2.5.14

#THE CHECK

The elastic drift ETABS reports is not the design drift.

$$\Delta = C_d \cdot \delta_{\text{elastic}} / I$$

Compare Δ against the allowable storey drift ratio, generally 0.020·h_{sx} for an ordinary building, 0.015 for occupancy category IV.

!Reading δ straight off the Story Drifts table and comparing it to 0.02 understates the drift by the factor C_d/I – five times for $C_d = 5$.

#WHEN IT FAILS, IN ORDER OF EFFECT

- 1 Wall LENGTH. Second moment of area goes as L^3 . Making a 2 m wall 3 m long multiplies its stiffness by 3.4. Making it thicker is linear.
- 2 Close the core. Four separate wall panels around a lift shaft are four weak cantilevers; the same four sharing corner joints are one box with many times the stiffness. They must share actual joints.
- 3 Insertion-point transformation. Drawing beams at the slab soffit and unticking "do not transform frame stiffness for offsets" makes the beam and slab act as a T-section. Worth 30-40% of the period.
- 4 Beam depth. Stiffness goes as d^3 , but a deeper beam only helps if the frame is carrying lateral load in the first place.
- 5 More columns at beam intersections – frame columns work in double curvature, $12EI/h^3$, not as full-height cantilevers.
- !6 Adding wall area at random. Check where the load actually goes first!
! (Results tab, wall share) before you add anything.

#TORSION

Read Story Max Over Avg Drifts. Over 1.2 is torsionally irregular and triggers the amplification of Sec 2.5.7.6.2; over 1.4 is extreme.
A first mode dominated by RZ means the stiffness centre is far from the mass centre – move wall, do not add it symmetrically.

7. Choosing R honestly

R is not a number you pick from the table you like. It follows from what the structure actually does, which you can only know after the analysis.

walls carry > 75% of base shear
bearing wall system, R = 5, if walls also carry the gravity
building frame system, R = 6, if a complete frame carries gravity
a DUAL system does NOT apply
walls carry 50-75%, frame takes 25% or more
dual system may be claimed – R = 6.5 with intermediate frames,
R = 7 with special frames – PROVIDED the frame is detailed for it
walls carry < 50%
frame dominated; check the walls are actually connected

!The dual-system definition requires the moment frame alone to resist at least 25% of the base shear. Claiming R = 6.5 when the walls take 92% is !a 30% under-design of the entire lateral system.

+The Results tab computes the share from pier forces. It needs pier labels +first, which is what the Piers tab is for.

8. Earth pressure on basement walls

#AT REST, because a propped basement wall does not move

$$K_0 = 1 - \sin \phi$$

below the water table use the buoyant unit weight and add the full hydrostatic pressure separately
add $K_0 \cdot q$ for any surcharge

Typical Dhaka values, but use YOUR geotechnical report:
 $\phi = 28-32^\circ$, $\gamma = 18 \text{ kN/m}^3$, $\gamma_{\text{sat}} = 20 \text{ kN/m}^3$, water table 1.5-2 m down.

#IN ETABS

There is no "lateral earth pressure" load type. Create a load pattern of type OTHER named H.

Apply it either as a joint-pattern-driven surface pressure, or panel by panel as a uniform pressure equal to the average over that panel's height, in the LOCAL 3 direction, sign chosen so it pushes inward.

!Check the sign by displaying the deformed shape under H alone. If the !basement wall bulges outward you have it backwards.

H then goes into basic combinations 2, 6 and 7 with a factor of 1.6.

!Where H resists the effect being checked – uplift, overturning, sliding – !use $0.9H$ if the earth is certain to be there, or drop it to zero.

9. Common modelling mistakes

!Beams stopping at the face of a wall with no joint there.
The beam ends in mid-air. Draw it to a wall joint, or align the wall mesh to it: Edit > Align Joints/Frames/Edges, Shift+Ctrl+M.

!Wall panels stacked but not sharing joints.
Two panels at the same line on adjacent storeys with different joints do not form one pier. Merge the joints first, then label the pier.

!One slab covering the whole floor with no beams under it.
Everything spans the full plan dimension and the beams need absurd depth. Put beams on every gridline the architecture allows.

!Auto edge constraint off.
A wall mesh node that does not coincide with a slab node transfers nothing. Assign > Shell > Auto Edge Constraint on all areas.

!Self weight counted twice.
Only one load pattern may have a self-weight multiplier of 1.0.

!Wind or seismic applied to a model with no diaphragms.
Auto lateral loads need diaphragms to find the exposure area and to place the storey forces. No diaphragm, no load.

!Design preferences out of step with the load pattern.
R, Ω , Cd and SDS live in two places. Changing the load pattern does not change the design preferences.

!Changing five things at once.
When it gets worse you will not know why. One change, re-run, record it.

10. Irregularity — Tables 6.1.3 and 6.1.4

An irregular building is not forbidden. It triggers extra requirements, and the requirement you most often trip over is that the equivalent static method stops being permitted.

#HORIZONTAL, Table 6.1.5

H1a torsional $\delta_{\text{max}} / \delta_{\text{avg}} > 1.2$

H1b extreme torsional $\delta_{\text{max}} / \delta_{\text{avg}} > 1.4$

H2 re-entrant corner a projection beyond 15% of the plan dimension

H3 diaphragm discontinuity an opening over 50% of the gross area, or a stiffness change over 50% between storeys

H4 out-of-plane offset of a vertical lateral element

H5 non-parallel systems

#VERTICAL, Table 6.1.4

V1a soft storey stiffness < 70% of the storey above, or < 80%

of the average of the three above
V1b extreme soft storey < 60%, or < 70% of the average of three
V2 mass irregularity mass > 150% of an adjacent storey
V3 vertical geometric dimension > 130% of an adjacent storey
V4 in-plane discontinuity of a lateral element
V5 weak storey storey strength < 80% of the storey above

#WHAT EACH ONE COSTS YOU
!H1 the accidental eccentricity must be amplified by
! $A_x = (\text{delta_max} / 1.2 \text{ delta_avg})^2$, capped at 3.0
! so the 5% eccentricity becomes 5%* A_x . In SDC D, H1b also amplifies
! diaphragm and collector forces by 25%.
!Vb is NOT PERMITTED in SDC D (Sec 1.5.4.3(b)); in SDC B or C it is
! allowed only up to two storeys or 9 m. BNBC 2020 has no SDC E or F.
!V2, V3, H2, H3 mean the equivalent static method may not be used - you
! have to run a response spectrum analysis.

+Tab 10 Checks screens all of these it can measure from the model, and
+tells you which ones need a hand measurement off the plan instead.

#THE FIX FOR TORSION IS NOT MORE WALL
Torsion comes from the stiffness centre sitting away from the mass centre.
Adding wall symmetrically does not move it. Adding wall on the FLEXIBLE
side does. Sometimes the right move is to remove or shorten a wall on the
stiff side.

11. P-Delta and spectrum scaling

#STABILITY COEFFICIENT Sec 2.5.7.9
 $\theta = P_x * \Delta / (V_x * h_{sx} * C_d)$
 P_x total gravity load at and above that storey (unfactored)
 Δ the DESIGN storey drift, $C_d * \delta / I$
 V_x the storey shear
 h_{sx} the storey height

$\theta \leq 0.10$ P-Delta may be ignored
 $\theta > 0.10$ include P-Delta, or multiply the drift by $1/(1-\theta)$
 $\theta > \theta_{max}$ not permitted
 $\theta_{max} = 0.5 / (\beta * C_d)$, never more than 0.25
 β is the ratio of demand to capacity in shear; 1.0 is safe

!A tall, slender, lightly braced building can pass the drift check and
!still fail θ . Check both.

#IN ETABS
Analyze > Set Load Cases to Run, or Define > Load Cases > Modify, and set
the P-Delta option. Use a P-Delta case built on 1.2D + 1.0L or the code's
gravity combination, not on the individual patterns.

#STOREY FORCE DISTRIBUTION Sec 2.5.7.6
 $F_x = V * w_x * h_x^k / \sum (w_i * h_i^k)$
 $k = 1.0$ for $T \leq 0.5$ s, 2.0 for $T \geq 2.5$ s, linear in between
Tab 10 computes this from the model's own storey masses. Compare it to the
ETABS Auto Seismic table. A disagreement means ETABS is using a different
 T , a different W , or a different k than you think it is.

#RESPONSE SPECTRUM SCALING Sec 2.5.9.4
A modal response spectrum analysis almost always gives a smaller base
shear than the static method, because the modal combination is a
statistical estimate.
!The RS base shear must be scaled UP to 85% of the static V .
scale factor = $0.85 * V_{static} / V_{dynamic}$
Apply it as the scale factor on the RS load case, not by editing results.
If the dynamic shear already exceeds 85%, no scaling is applied - you do
not scale down.

12. Foundations and service loads

#TWO DIFFERENT SETS OF COMBINATIONS, FOR TWO DIFFERENT JOBS

STRENGTH (UD / UW / UE, Sec 2.7.3.1)

sizes the reinforcement and checks the concrete

SERVICE / ASD (SD / SW / SE, Sec 2.7.2.1)

sizes the footing plan area against the allowable bearing pressure,
and checks settlement, overturning and sliding

!Using factored loads to size a footing oversizes it by roughly 50%.

!Using service loads to design the footing concrete under-reinforces it.

!They are not interchangeable and you need both in the model.

#THE ASD COMBINATIONS

1 D

2 D + L

3 D + (Lr or R)

4 D + 0.75L + 0.75(Lr or R)

5 D + (0.6W or 0.7E)

6 D + 0.75(0.6W or 0.7E) + 0.75L + 0.75(Lr or R)

7 0.6D + 0.6W + H

8 0.6D + 0.7E + H

E is 0.7E for ASD, and Ev rides with it, so the dead factor becomes

1 + 0.7(0.2 SDS) additive and 0.6 - 0.7(0.2 SDS) counteracting.

+Tick "also build SERVICE combinations" on tab 7 and they are created

+alongside the strength set, and deliberately NOT registered as concrete

+design combinations.

#WHAT TO HAND THE FOUNDATION DESIGNER

Tab 12 exports every restrained joint under every combination, plus an
envelope per joint: FZ max, FZ min, maximum shear, maximum moment, and a
flag for any joint that goes into uplift.

!Uplift almost always appears under the 0.9D combinations. That is exactly

!what those combinations are for. A footing in uplift needs a tie beam,

!more dead weight, or piles - it is not a modelling error.

#ALLOWABLE BEARING

Take it from your geotechnical report, at the depth you are actually
founding at, and check the settlement criterion separately - a pressure
that satisfies bearing capacity can still settle too much on soft Dhaka
clay.

#THE PRELIMINARY PAD SIZER

Plan area comes from the SERVICE envelope, $e = M/P$, against q_{allow} .

While $e \leq L/6$ the pressure is the trapezoid $q = P/A (1 + 6e/L)$; past
that the base lifts and the partial-contact form $q = 2P/(3B(L/2 - e))$
governs - the trapezoid under-predicts it, so both are in there.

+Tick "rectangular pad where $e > B/6$ " and an eccentric pad grows its
+long side along the moment until the whole base bears, up to the L/B
+cap you set, instead of growing a square in both directions.

Depth is the larger of two-way punching on the perimeter at $d/2$ and
one-way (beam) shear at d from the column face, both on the FACTORED
peak pressure; the sheet says which one governed each pad, what
bearing pressure it actually reaches, and - where the base still
lifts - how much of it is left in contact with the soil.

!The resultant moment is applied about one axis. Biaxial moment,

!combined, strap and mat footings, sliding and settlement are not in

!here - that work belongs in SAFE.

13. What this program does not do

This is a workflow assistant. It reads your model, computes BNBC 2020
parameters and writes patterns, loads, combinations and pier labels.

It does NOT:

- save your model – every action leaves the file unsaved so you can undo
- run the analysis – press F5 in ETABS yourself
- design anything – ETABS does that
- decide R for you – it tells you what the wall share implies
- replace your judgement or your professional seal

!Every number it prints carries a clause reference. Check them against
!your printed copy of BNBC 2020 Part 6 Chapter 2 before you rely on them.
!Where this program and the code disagree, the code is right.

#IF SOMETHING GOES WRONG
Nothing here saves the model. Close ETABS without saving and you are back
where you started. That is deliberate.

Outputs and files

Output	Where it comes from
Design-basis report (PDF)	Report
Foundation reactions, quantity take-off, member schedule	Output
Project file (.aex) — your inputs, reusable	Output
AutoCAD detail sheets (DXF) + bar bending schedule	Detailing
Spreadsheet exports (.xlsx, or CSV without openpyxl)	several tabs

The detail sheets

Drawn 1:1 in millimetres on A0, A1, A2 or A3 at the plot scale you choose, with text and the title block scaled so that 2.5 mm type comes off the plotter at 2.5 mm. Separate layers for concrete, rebar, links, dimensions, text and the schedule.

The bar bending schedule is accumulated **while the drawings are produced**, so the schedule and the drawings cannot disagree with each other. It carries bar mark, diameter, number, cut length, shape and weight, totalled by diameter and split by member family — beam, column, slab, wall.

Two things are asked for and never guessed, because no model file records them and both change the steel:

- **The seismic detailing class** — ordinary, intermediate or special —

which sets the tie spacing under Sec 8.1.9.

- **The coarse aggregate.** Brick aggregate needs **1.5 times** the

shrinkage and temperature steel under Sec 8.1.11.2(b). There is no ACI equivalent of that clause, and a slab detailed for stone but poured in brick is short of steel.

Troubleshooting

Symptom	Cause and cure
Connect does nothing	Elevation mismatch. Run both as administrator, or neither.
Attaches to the wrong model	More than one ETABS instance. Close the extras.
"No modal results"	The analysis has not been run, or has no modal case. Run it in ETABS.
"No beam design results"	Run the concrete design in ETABS. EtabsX reads its results; it does not design members.
Export produced a CSV, not xlsx	openpyxl is not installed. Install it, or use the CSV — the data is identical.
First launch is slow, two processes	Normal for a one-file build; it unpacks to a temp folder on start.

Symptom

Cause and cure

"Access has ended"

The free period or the licence has run out. Send the machine code shown to TwinAnalytic and paste the key back into the same box.

A key is refused

It belongs to a different computer, or it has expired. The message says which.

Windows blocks the download

The file is not code-signed. More info → Run anyway, or check the published SHA-256.

Limits and disclaimer

EtabsX implements a defined subset of BNBC 2020. It does not implement the Code entire, and the **Guide** tab in the program lists what it deliberately leaves alone — including time-history analysis, components and cladding, site-specific studies for site classes S1 and S2, capacity-design shear from probable moments, and shear-wall boundary elements on demand.

Every number it prints carries its clause so that you can check it. That is the intended use: check the arithmetic you would otherwise do by hand, not replace the engineer who is accountable for the result.

This program does not certify anything. The design, the drawings and the seal are the responsibility of the engineer who signs them.

EtabsX by TwinAnalytic · <https://twinanalytic.com/etabsx>